Jimin Go

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Research Interests

Advanced Packaging Modeling, Electronic Design Automation, Physics-Informed Machine Learning

Education

Arizona State University, AZ, USA — Aug. 2025 – Present

- Doctor of Philosophy in Computer Engineering
- GPA: 4.0 out of 4.0
- Advisor: Leslie. K. Hwang
- Awarded University Graduate Fellowship for research progress and strong academic performance

Weber State University, UT, USA — Jan. 2024 – May. 2025

- Bachelor of Science in Management Information Systems
- GPA: 3.92 out of 4.0
- Top 10 percent of undergraduate students in the Goddard School of Business & Economics

Incheon National University, Incheon, Korea — Mar. 2021 – Aug. 2025

- Bachelor of Engineering in Industrial and Management
- GPA: 4.41 out of 4.5
- Recipient of Presidential Award (Ranked 1st in the College of Engineering)

Research Experiences

Independent Researcher, Arizona State University — Mar. 2026 – Present

- Research goal: Developing a physics-informed machine learning framework for autonomous microchannel geometry generation, targeting thermal management optimization in advanced packaging (2.5D SiP, 3D-IC)
- Developed a PyFluent scripting pipeline to fully automate Ansys CFD simulation execution, replacing manual Ansys workflows and enabling large-scale training data generation for ML models
- Designing a diffusion model and Fourier Neural Operator (FNO) to learn the mapping between thermal boundary conditions and optimal microchannel geometries without manual design iteration

Graduate Research Assistant – SHIELD-USA Project, Arizona State University — Aug. 2025 – Present

- Led all monitor designs for the SHIELD-USA demonstrator RDL Fan-out test vehicle, covering parametric floorplanning, bump planning, daisy chain generation, and layer-aware autorouting using Synopsys 3DIC Compiler

- Investigated autorouter failure modes under advanced packaging geometries and implemented staged routing, exit angle control, and routing order optimizations to resolve them
- Diagnosed and resolved DRC/LVS violations in IC Validator (ICV), reducing violation count from >100,000 to <30,000 by identifying false positives from geometric vectorization, precision scaling mismatches, and runset logic errors
- Contributing to the construction of the Process/Assembly Design Kit (PDK/ADK) for SHIELD packaging technologies, bridging IC-style EDA flows with advanced packaging design rules
- Developing post-route static timing analysis workflow integrating StarRC parasitic extraction and Synopsys PrimeTime signoff

Undergraduate Research Assistant, Weber State University May. 2024 – Sep. 2025

- Investigated generative AI and computer vision-based defect detection for real-world second-hand e-commerce platforms
- Built a full-stack web prototype (Django, HTML/CSS, JavaScript) that auto-generates product annotations and flags defects from seller-uploaded photos using a YOLO-based pipeline

Undergraduate Research Assistant, Incheon National University Nov. 2022 – Dec. 2023

- Designed a discrete-event, constraint-aware scheduling engine for cyclic electroplating lines, replacing days-long manual simulation with automated feasibility analysis
- Developed a backtracking-based optimization algorithm with cyclic feasibility enforcement, achieving 25% reduction in carrier count and 1.5% decrease in cycle time while maintaining 100% schedule consistency

Publications

2026 **ElectroTwin: A digital twin-driven framework for designing and scheduling in electroplating facilities**
Donghun Lee, Taehyun Noh, Jimin Go, In-Beom Park, Kwanho Kim
International Journal of Precision Engineering and Manufacturing

SHIELD-USA: Leap-ahead Organic Substrate Enabled by Fan-out Technology, Domestic Materials, Process Innovations, and Novel EDA Workflows
Georgios Dogiamis, Tim Takeuchi, Craig Bishop, Leslie Hwang, Alec Bunn, Jimin Go, M.Magnavita, M. Naujokaitis, K. Larsen, Jason Upton, Tom McLaughlin, Anwar Mohammad, V. Singaram, Robert Freeman
Conference on Government Microcircuit Applications & Critical Technology (GOMACTech)

2025 **Multi-feature representation-based graph attention networks for predicting potential supply relationships in a large-scale supply chain network**
Donghun Lee, Jimin Go, Taehyun Noh, Seookwoo Song
Expert Systems with Applications

Autonomous annotations system for second-hand E-commerce platform using Generative Artificial Intelligence (Extended version of 2024 paper)
Jimin Go, Taehyun Noh, Donghun Lee, Seokwoo Song
Workshop on the Korean Chapter of the Association for Information Systems (KrAIS)

A study on a method for deciding optimized the number and moving range of hoists
Jimin Go, Taehyun Noh, Haeun Lee, Kwanho Kim
Conference on Society for e-business studies (SEBS)

2024 **Autonomous annotations for second-hand E-commerce platforms using Generative Artificial Intelligence**
Jimin Go, Taehyun Noh, Seokwoo Song
Symposium on 2024 Fall Research and Engagement at Weber State University

2023 **A dynamic solution platform for multi-building cooling operation**
Tackkyoon Woo, Suhee Kim, Jimin Go, Taehyun Noh, Kwanho Kim
Conference on the Korean Institute of Industrial Engineers (KIIE)

Honors and Awards

2026 University Graduate Fellowship, Arizona State University
2025 Presidential Award, Incheon National University
Goodard Scholar, Weber State University
2024 Information Systems Game Day Analytics Competition, University of Utah
Engineering College Early Taste Early Decision Competition, Incheon National University
Employment-linked Matrix Scholarship, Incheon National University
2023 The 19th Industrial Engineering Project Competition, Korean Institute of Industrial Engineers
Academic Excellence Scholarship, Incheon National University
2022 Academic Excellence Scholarship, Incheon National University
2021 Academic Excellence Scholarship, Incheon National University

Patent

2024 Kwanho Kim, Jimin Go, Taehyun Noh. 2024. Cyclic hoist scheduling method, manufacturing method, and cyclic hoist scheduling apparatus using the same. Korean Patent No. 10-2024-0007227.

Academic Service

2026 Secondary Reviewer, Design Automation Conference (DAC)

Teaching Experiences

Teaching Assistant

2026 EEE 425/591: Digital Systems and Circuits, Arizona State University
Undergraduate and Graduate level course, 100+ students (Instructor: Leslie Hwang)

2025 MIS 2010: Business Computer Skills, Weber State University
Undergraduate level course, 50+ students

MIS 2030: Introduction to Business Analytics, Weber State University
Undergraduate level course, 50+ students (Instructor: Seokwoo Song)

MIS 2040: Business Analytics with Python, Weber State University
Undergraduate level course, 50+ students (Instructor: Taewon Le)

Technical Skills

Packaging & Physical Design	RDL Fan-Out Design, Floorplanning, PDK/ADK development, DRC/LVS check
EDA Tools	Synopsys 3DIC Compiler, IC Validator (ICV), StarRC
Simulation	Ansys CFD, PyFluent
Hardware Description	Verilog
Programming Languages	Python, Tcl, SQL
ML Frameworks	TensorFlow, PyTorch, NumPy, Pandas, Scikit-learn, YOLO, OpenCV, Graph Neural Network, FNO, and many others
Web Development	HTML, CSS, JavaScript, Django
Operating Systems	Windows, Linux, Unix
Tools	Git, Tableau, KNIME